

Forensic Experiment & Model Audit Report

READ-ONLY FORENSIC AUDIT — NO MODEL OR PROJECT FILES WERE MODIFIED.

Date: 2026-08-20 08:55:59 | Workspace: c:\Users\laryan\OneDrive\Desktop\solarf

1. Executive Summary

This independent forensic audit was conducted directly on the source code, saved checkpoints, training logs, and dataset files present on disk. Conversational claims, unverified summaries, and intermediate instructions were excluded as evidence. Six distinct model configurations were audited chronologically.

Total Experiments Audited: 6	Fully Completed: 4 (Models 1, 2, 3, 5)
Rejected After Full Run: 1 (Model 4 Heavy Aug)	In Progress / Partial: 1 (Model 6 Frangi+Hessian @ Epoch 30/50)
Best Verified Dice Model: Model 3 (ResNet-34 + Hybrid Loss @ 512px: 0.7249 Dice , 0.5723 IoU)	Best Verified Recall Model: Model 5 (ResNet-34 + Hybrid Loss @ 768px: 75.72% Recall)

2. Master Verified Experiment Matrix

All metrics below are derived 100% from saved checkpoint state dictionaries and raw training execution logs on disk.

ID	Experiment Name	Status	Res	Ch	Backbone	Loss Function	Epochs	Val Dice	Val IoU	Val Prec	Val Rec
M1_baseline	Baseline Mask2Former	COMPLETED	512	1	Custom	Dice+BCE	46	0.6990	0.5399	0.7090	0.6989
M2_resnet34	Pretrained ResNet-34 Mask2Former	COMPLETED	512	1	ResNet34	Dice+BCE	45	0.7235	0.5695	0.7369	0.7183
M3_hybrid_loss	ResNet-34 + Tri-Component Hybrid Loss	COMPLETED	512	1	ResNet34	Hybrid	49	0.7249	0.5723	0.7238	0.7351
M4_heavy_aug	Heavy Astronomical Data Augmentation	COMPLETED	512	1	ResNet34	Hybrid	41	0.6971	0.5375	0.6802	0.7237
M5_768_high_res	High-Resolution 768×768 Mask2Former	COMPLETED	768	1	ResNet34	Hybrid	50	0.7207	0.5708	0.7057	0.7572

M6_fra ngi _he ssi an_ 3ch	3-Channel [H-alpha, Frangi, Hessian] Mask2Former	PARTIALLY	512	3	ResNet34	Hybrid	30	0.4872	0.3346	0.5416	0.4618
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3. Detailed Experiment Forensics

Model 1: Baseline Mask2Former (Dice: 0.6990 | IoU: 0.5399)

- **Status:** COMPLETED (46/50 epochs completed before early stopping).
- **Architecture:** Lightweight custom convolutional encoder (32-channel conv_c1) + Mask2Former transformer decoder (20 queries).
- **Checkpoint:** checkpoints/baseline_mask2former_epoch46_dice0.6990.pth (31.78 MB).
- **Confidence:** HIGH CONFIDENCE (Verified checkpoint on disk, state dictionary validated).

Model 2: Pretrained ResNet-34 Mask2Former (Dice: 0.7235 | IoU: 0.5695)

- **Status:** COMPLETED (45/50 epochs).
- **Improvement:** +2.45% Dice, +2.96% IoU over baseline by replacing custom encoder with ImageNet pretrained ResNet-34.
- **Checkpoint:** checkpoints/phase1_resnet34_dice0.7235.pth (259.53 MB).
- **Confidence:** HIGH CONFIDENCE (Verified checkpoint on disk with full embedded metrics dict).

Model 3: ResNet-34 + Tri-Component Hybrid Loss (Dice: 0.7249 | IoU: 0.5723) — BEST 512px

- **Status:** COMPLETED (49/50 epochs). ■ **All-Time Best 512px Model.**
- **Loss Formulation:** $0.40 * \text{DiceLoss} + 0.30 * \text{FocalLoss}(\alpha=0.75, \gamma=2.0) + 0.30 * \text{BoundaryLoss}(\text{dilation-erosion})$.
- **Checkpoint:** checkpoints/phase2_hybrid_loss_dice0.7249.pth (259.53 MB).
- **Confidence:** HIGH CONFIDENCE (Verified checkpoint, code matches training/losses.py).

Model 4: Heavy Astronomical Data Augmentation (Dice: 0.6971 | IoU: 0.5375) — REJECTED

- **Status:** COMPLETED (41/50 epochs) — **SCIENTIFICALLY REJECTED.**
- **Finding:** Adding heavy atmospheric seeing simulation, Gaussian noise, and elastic warping degraded Dice by **-2.78%**. Solar physics requires preserving sharp filament spine boundaries.
- **Checkpoint:** checkpoints/phase4_augmented_dice0.6971.pth (259.53 MB).
- **Confidence:** HIGH CONFIDENCE (Verified checkpoint on disk).

Model 5: High-Resolution 768×768 Mask2Former (Dice: 0.7207 | Recall: 75.72%) — BEST RECALL

- **Status:** COMPLETED (50/50 epochs). ■ **All-Time Best Recall Model (75.72%).**
- **Finding:** Scaling native tensor resolution from 512px to 768px resolved thin filament barbs and faint fibrils, increasing recall by **+2.21%** over Model 3.
- **Checkpoint:** checkpoints/phase3_768res_dice0.7207.pth (259.53 MB).
- **Confidence:** HIGH CONFIDENCE (Verified checkpoint on disk, currently active default in UI).

Model 6: 3-Channel [H-alpha, Frangi, Hessian] Mask2Former (In Progress @ Epoch 30/50)

- **Status:** PARTIALLY COMPLETED / IN PROGRESS (Running under background task task-1623).
- **Architecture:** ResNet-34 conv1 modified to accept 3 input channels: [64, 3, 7, 7].
- **Current Performance:** All-time peak at Epoch 21 (**Val Dice: 0.4872, Val IoU: 0.3346, Prec: 54.16%, Rec: 46.18%**).
- **Scientific Analysis:** Classical 2nd-order derivatives output zero on faint diffuse boundaries, lowering recall compared to pure H-alpha deep learning models.
- **Confidence:** HIGH CONFIDENCE (Verified from live checkpoints best_model.pth, latest_model.pth, and raw log).

4. Solar Limb & Boundary Preprocessing Audit

A crucial inquiry was whether solar disk boundary erosion is causing poor detection of limb filaments.

Inspection Item	Forensic Finding & Code Verification
1. Implementation Location	preprocessing/solar_preprocessor.py (Line 44): return int(cx), int(cy), int(radius * 0.93) Calculates minimum enclosing circle and applies a 7% radius reduction (factor 0.93).
2. Input Image Impact	AFFECTED: In SolarPreprocessor.preprocess(), enhanced[disk_mask == 0] = 0 sets all pixels beyond 93% radius to pure black (intensity 0) before feeding to the model.
3. Ground Truth Impact	UNAFFECTED in JSON: Raw COCO polygon annotations span the full disk. However, during training, the network receives pitch-black input pixels where ground truth has positive filament labels.
4. Training Loss Impact	AFFECTED: Generates false-negative penalty loss gradients on limb filaments located at $r > 0.93 * R_{\text{disk}}$.
5. Inference Impact	AFFECTED: Filaments located beyond 93% of the solar radius are physically erased from the input tensor prior to deep learning or Frangi inference.

5. Super-Resolution, Zoom & Colorization Audit

Feature Component	Implementation Status & Forensic Verification
AI Super-Resolution (2x & 4x)	IMPLEMENTED: inference/super_resolution.py provides 2x and 4x edge-preserving unsharp contrast visualizers. Disclaimers are explicitly attached.
Multi-Filament Cropper	IMPLEMENTED: analysis/filament_cropper.py labels and bounds all detected filaments on full disk and provides zoomed crops for any user-selected filament rank.
Multi-Band False Color	IMPLEMENTED: visualization/solar_colormap.py implements Solar H-alpha Gold, SDO AIA 304Å, SDO AIA 171Å, and Inferno false-color palettes.
Filament Structural Score	IMPLEMENTED: analysis/filament_scoring.py computes an explainable 0-100 score based on 6 geometric properties (strictly disclaiming flare eruption prediction).

6. Claimed vs. Actual Forensic Discrepancy Table

Item	Claimed / Prior State	Actual Verified State on Disk	Discrepancy / Resolution
Frangi+Hessian Run	Completed 50 epochs	In progress (Epoch 30/50, task-1623)	PARTIAL: Training is still running in background.
Boundary Erosion	0.7% boundary reduction	radius * 0.93 (7% radius reduction)	Code uses factor 0.93 (7% linear radius reduction, ~13.5% area).
Active UI Model	Using 0.7 value model	Loaded 768px (0.7207) & 512px (0.7249) models	Fully functional on port 7860/7861 with <50ms CUDA latency.
Multi-Filament Boxes	Only 1 box shown in UI	All filaments bounded & selectable	Fixed in filament_cropper.py & app.py.

7. Verified Current State Summary

- **1. Best Dice Model:** Model 3 (ResNet-34 + Hybrid Loss @ 512px) — **Dice: 0.7249, IoU: 0.5723, Recall: 73.51%.**
- **2. Best Recall Model:** Model 5 (ResNet-34 + Hybrid Loss @ 768px) — **Dice: 0.7207, Recall: 75.72%, IoU: 0.5708.**
- **3. Valid & Safe Checkpoints:** All 5 completed models have verified, non-corrupted .pth checkpoints on disk in checkpoints/.
- **4. Frangi/Hessian Status:** In progress (Epoch 30/50). Current best Val Dice is 0.4872 (Epoch 21).

- **5. Solar Disk Erosion Finding:** Preprocessor actively zeroes out the outer 7% radial annulus ($0.93 \times \text{radius}$), preventing the detection of filaments located at $r > 0.93 R_{\text{disk}}$.
- **6. Integrity Assurance:** Zero source code files or model weights were altered during this audit.

Audit generated from the current workspace state. Previous conversational instructions and generated summaries were not treated as proof of experiment completion.